Theory

(a) What are the limitations of Classical optimization problems? How Non Linear Programming (NLP) tries to overcome these problems?

(b) What is Kuhn-Tucker conditions (KTC)? Derive and interpret the KTC for a simple NLP. [Broad question]

(c) "Cusps are the most frequently cited culprits for the failure of the Kuhn-Tucker conditions, but the truth is that the presence of a cusp is neither necessary nor sufficient to cause those conditions to fail at an optimal solution". Explain this statement with appropriate example. [Note: Chiang 13.2; Irregularities at Boundary Points with example 1, 2, and 3. This is a broad question]

(d) Show that the constraint qualification is satisfied if, for any point x* on the boundary of the feasible region, there exists a qualifying arc for every test vector dx. [The Constraint Qualification, page 415]

Math

(a) Apply the Kuhn-tucker conditions to solve the following problem:

$$\text{Maximize } f(x, y) = xy$$
$$\text{Subject to } x^2 + y^2 \leq 1$$
$$\text{And } x, y \geq 0$$

(b) Apply the Kuhn-tucker conditions to solve the following problem:

$$\text{Maximize } Z = 2x - y^2$$
$$\text{Subject to } 2x^2 + 2y^2 \leq 2$$
$$\text{And } x, y \geq 0$$

(c) Solve the following cost minimization problem

$$\text{Minimize } C = (x_1-3)^2 + (x_2-4)^2$$
$$\text{Subject to } x_1 + x_2 \geq 4$$
$$\text{And } x_1, x_2 \geq 0$$

(d) A consumer lives on an island where she produces two goods, x and y, according to the production possibility frontier $x^2 + y^2 \leq 200$, and she consumes all the goods herself. Her utility function is:

$$U = xy^3$$

The consumer also faces an environmental constraint on her total output of both goods. The environmental constraint is given by $x + y \leq 20$.

(i) Write out the Kuhn-Tucker conditions.

(ii) Find the consumer's optimal x and y. Identify which constraints are binding.

8. a) Typically during times of war the civilian population is subject to some form of rationing of basic consumer goods. Consider the case of a two-good world where both goods, x and y. are rationed. Let the consumer's utility function be U = U(x, y). The consumer has a fixed money budget of B and faces the money prices Px and Py. Further, the consumer has an allotment of coupons, denoted C, which can be used to purchase both x or y at a coupon price of cx and cy. 10

i) Formulate the consumer's utility maximization problem under the non-linear programming framework and derive the Kuhn-Tucker conditions (KTC).

Let's suppose the utility function is of the form $U = xy^2$. Further, let B =100, $p_x = p_y = 1$ while C =120 and $c_x = 2 \; and \; c_y = 1$.

ii) Find the amount of the commodities at which utility will be maximized.

b) Solve the following expenditure minimization problem 4

$$\begin{aligned} &Minimize \quad Z = 2y - x^2 \\ &subject\ to \ x^2 + y^2 \le 1 \\ &and \qquad x, y \ge 0 \end{aligned}$$